



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-10: 2 Classes



*Please listen Carefully
&*

Keep your phone(s) on silent mode

Diversity and relatedness



How are they related ?

- Composed of matter and energy.

How are they different ?

- One can replicate, another can't.



How are they related ?

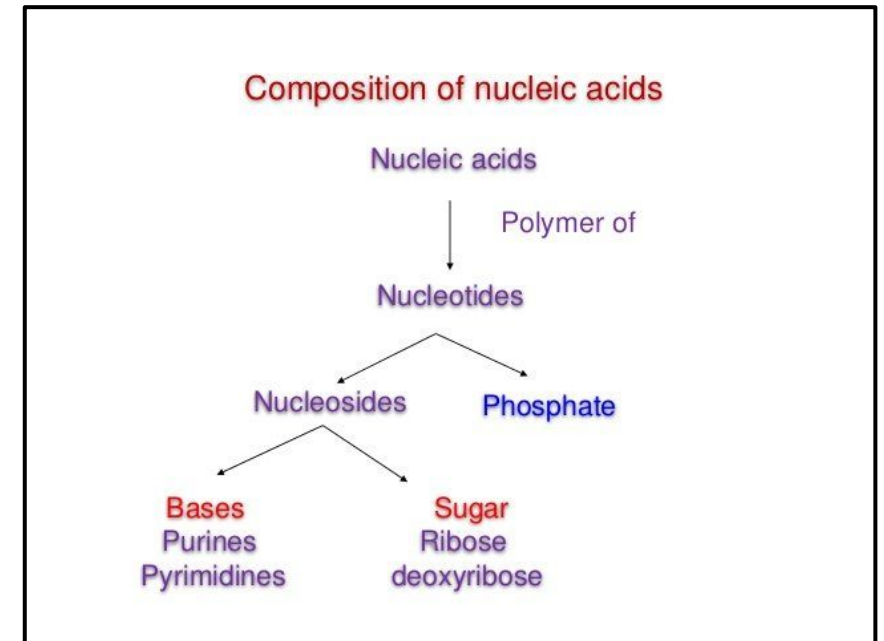
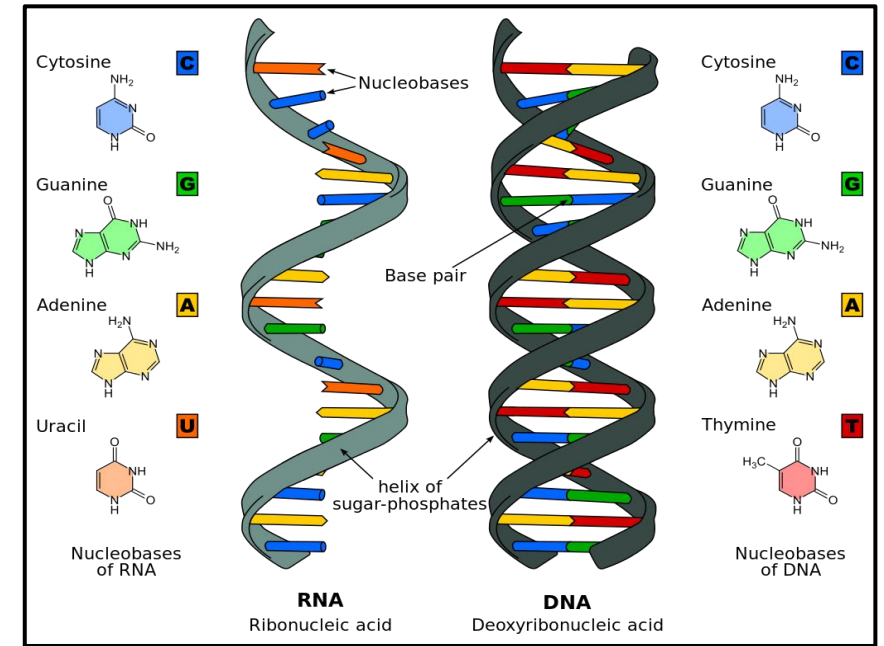
- All can replicate

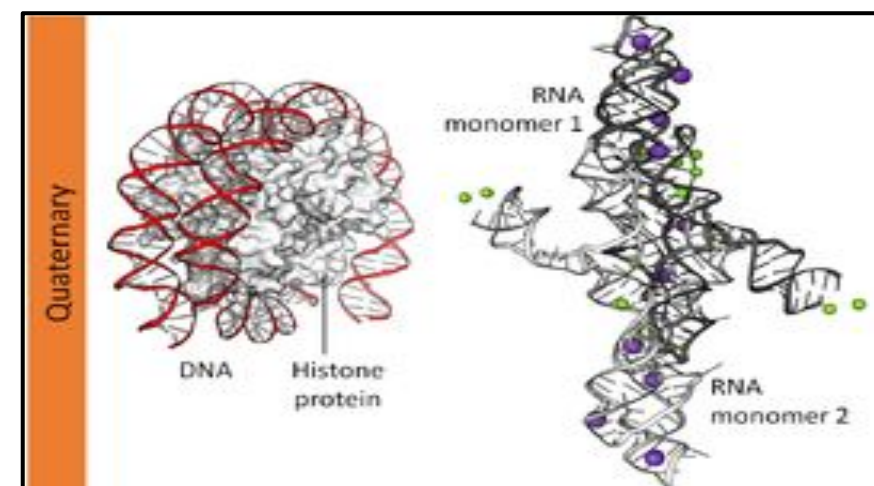
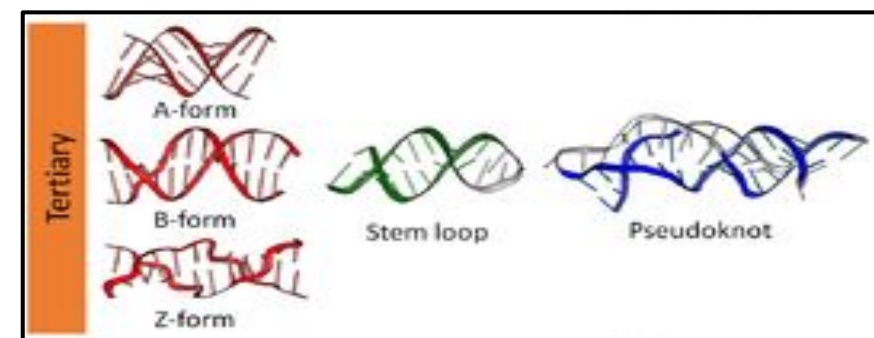
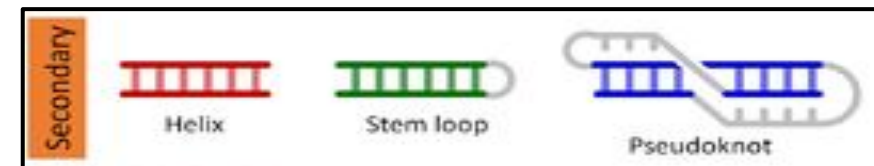
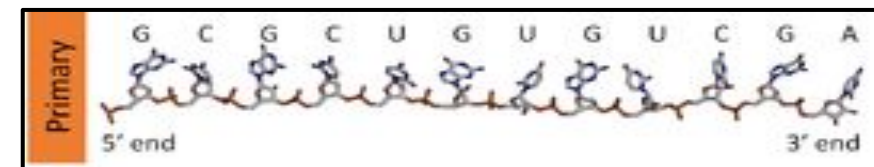
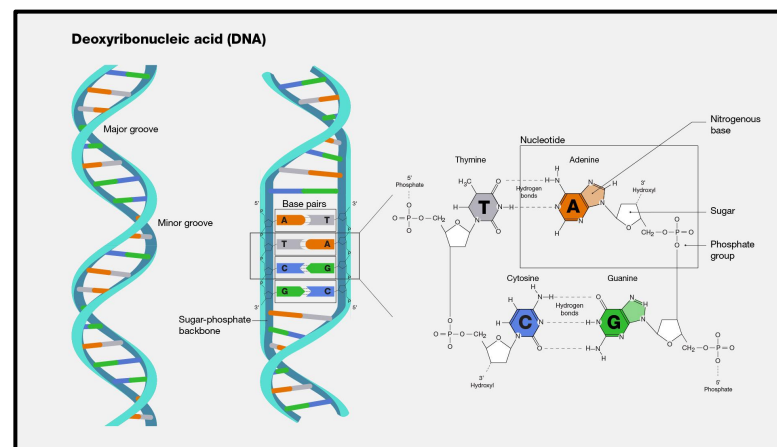
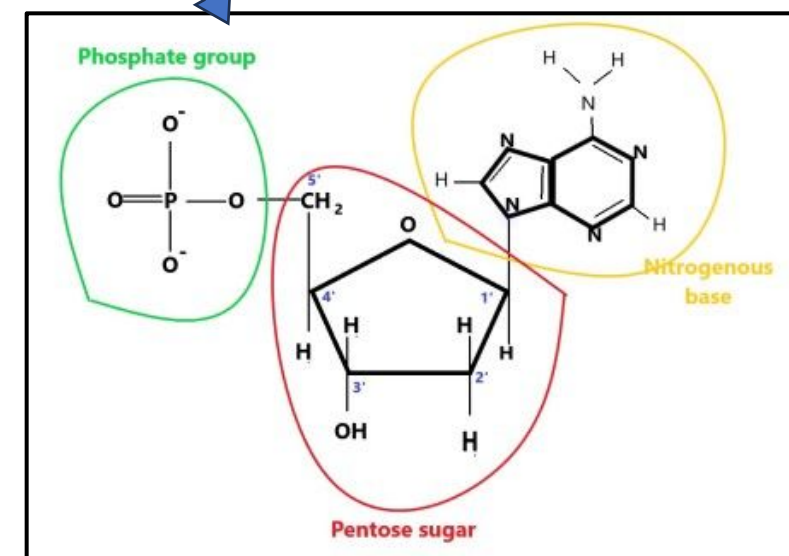
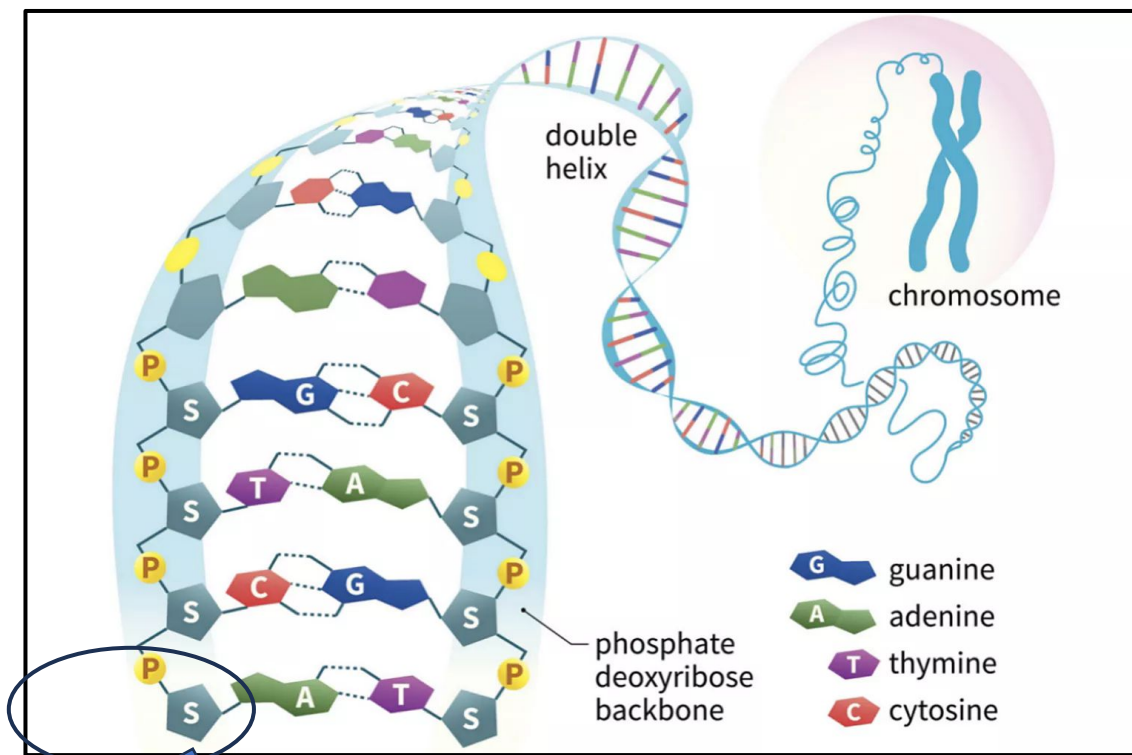
How are they different ?

- Genetically diverse.

Nucleic acids: structure and composition

- Biomolecules that **store and transmit genetic information** in living organisms.
- Include **DNA and RNA**.
- Composed of monomers called **nucleotides**.
- Nucleotides consist of a **sugar molecule**, a **phosphate group**, and a **nitrogenous base**.
- Encode the **genetic instructions** that determine an organism's traits and functions.
- Nucleic acids **can replicate themselves** to ensure the accurate transmission of genetic information.

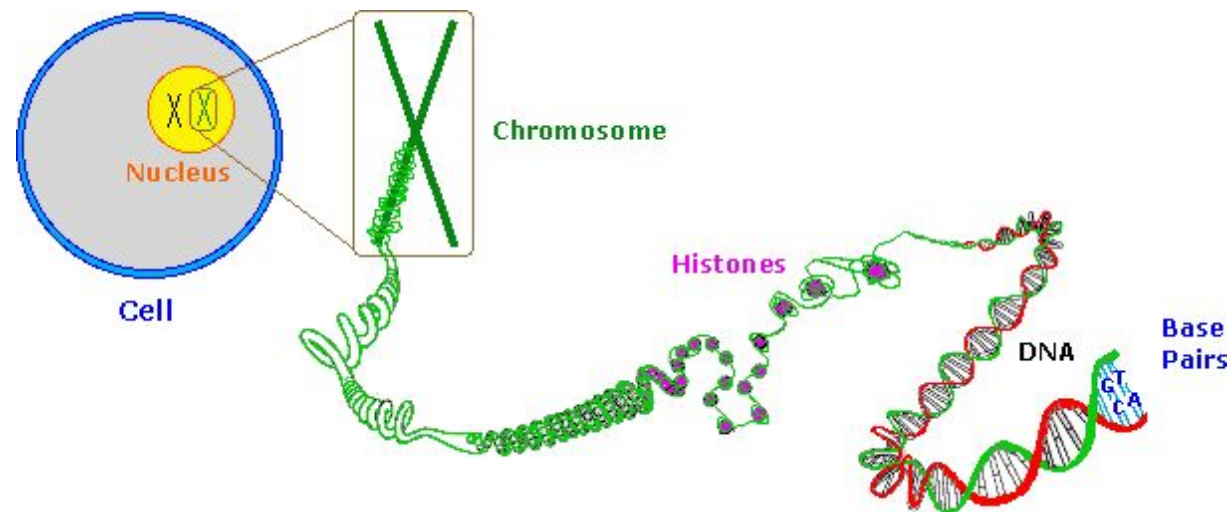
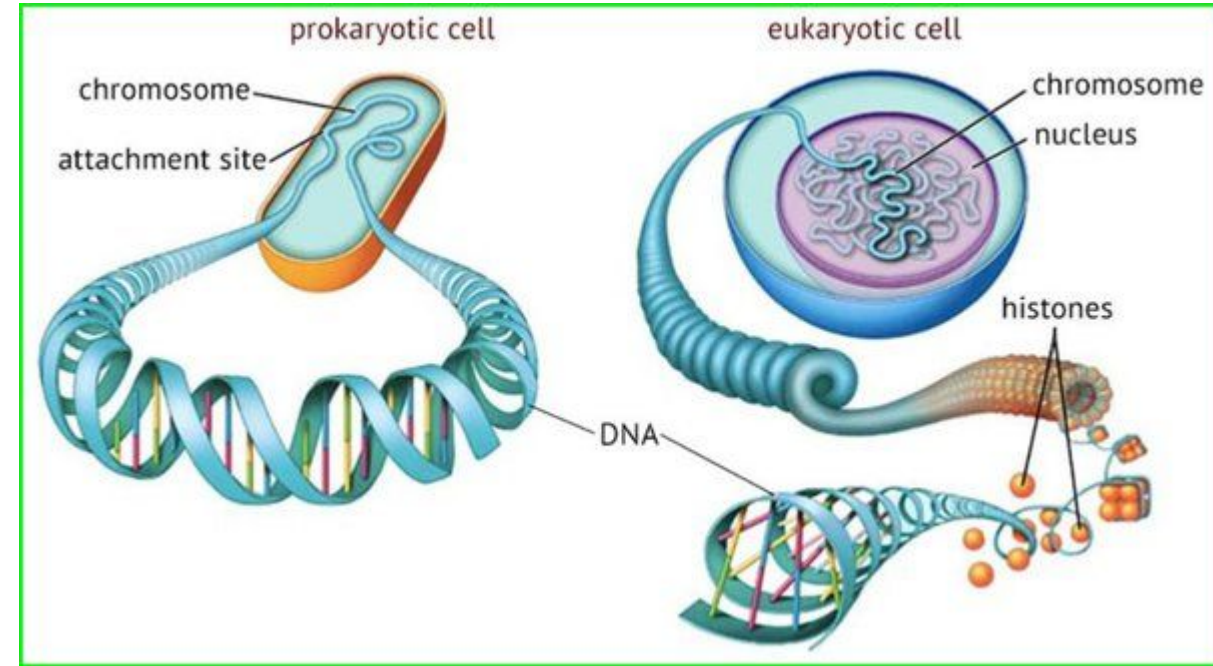




Chromosomes and DNA



- Most bacteria have a **single, circular set (nucleoid)** that contains majority of genes and is attached to the cell membrane.
- In addition to the main chromosome, bacteria may carry **extrachromosomal elements called plasmids (a small circular DNA molecule found in bacteria and some other microscopic organisms. Plasmids are physically separate from chromosomal DNA and replicate independently)**.
- Chromosomes are thread-like structures composed of **DNA and associated proteins** found in the nucleus of eukaryotic cells.
- The **number of chromosomes** in an organism's cells is **specific to its species**.
- Each chromosome consists of DNA wrapped around **histone proteins**. This coiling helps compact the genetic material and ensures it can be properly organized during cell division



Types of DNA

A-DNA:

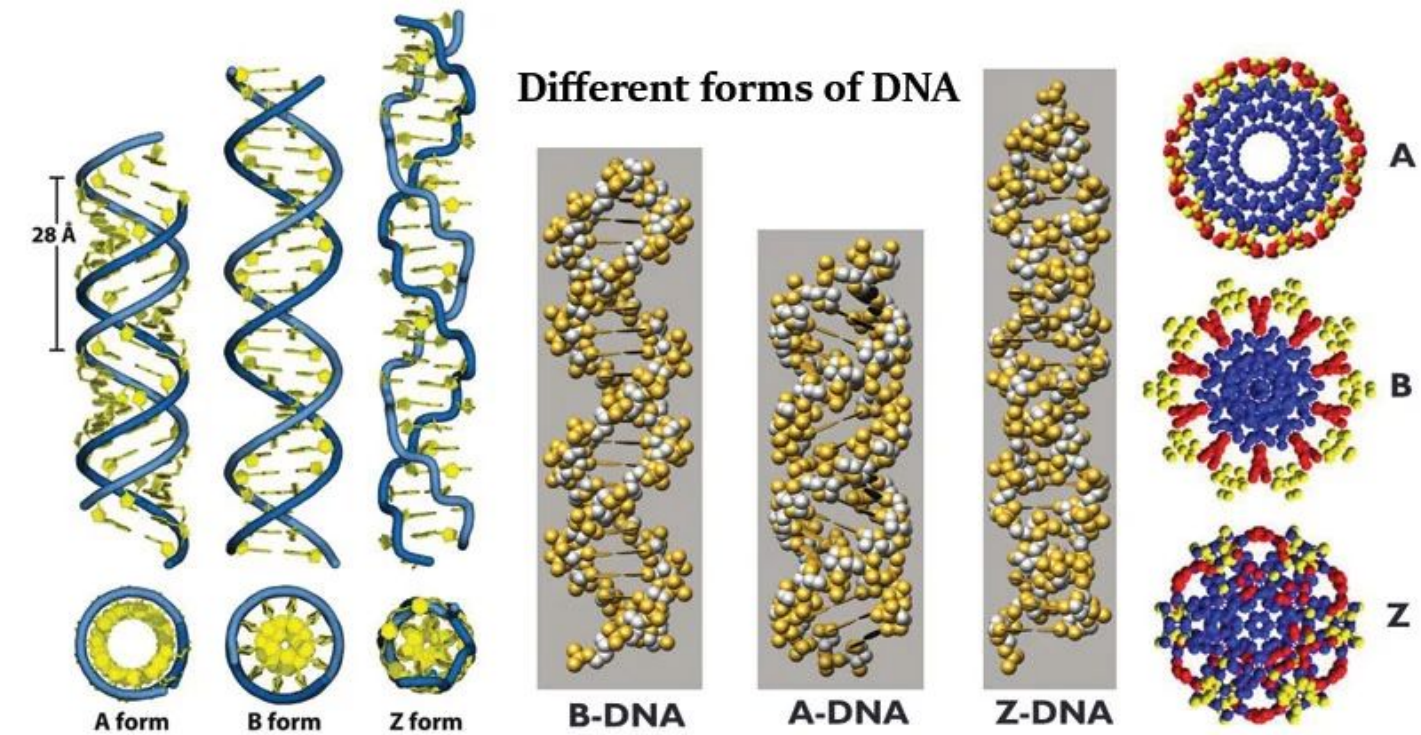
Conditions: Dehydrated DNA samples or under certain experimental conditions, such as **high salt concentrations**.
Biological Role: Less common in biological systems, it may play a role in **specific DNA-protein interactions and DNA packaging**.

B-DNA:

Structure: Well-known form of DNA.
Right-handed helical structure with 10 base pairs per turn.
Biological Role: B-DNA is the **primary conformation** of DNA. Serves as the **template for DNA replication, transcription, and gene expression**.

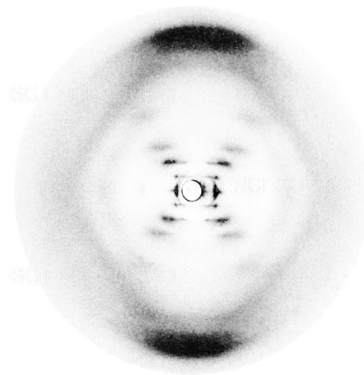
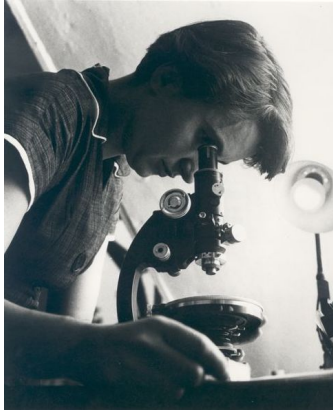
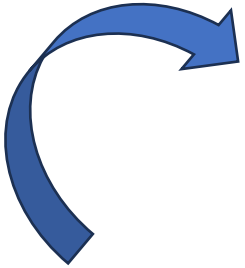
Z-DNA:

Conditions: **High salt concentrations, negative supercoiling**, and specific DNA sequences containing **alternating purine-pyrimidine bases**.
Biological Role: Z-DNA is less common in biological systems but has been **implicated in gene regulation, particularly in transcription initiation and chromatin structure**.

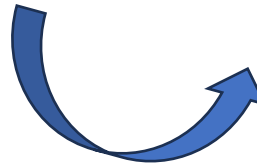


Feature	B-DNA	A-DNA	Z-DNA
Type of helix	Right-handed	Right-handed	Left-handed
Helical diameter (nm)	2.37	2.55	1.84
Rise per base pair (nm)	0.34	0.29	0.37
Distance per complete turn (pitch) (nm)	3.4	3.2	4.5
Number of base pairs per complete turn	10	11	12
Topology of major groove	Wide, deep	Narrow, deep	Flat
Topology of minor groove	Narrow, shallow	Broad, shallow	Narrow, deep

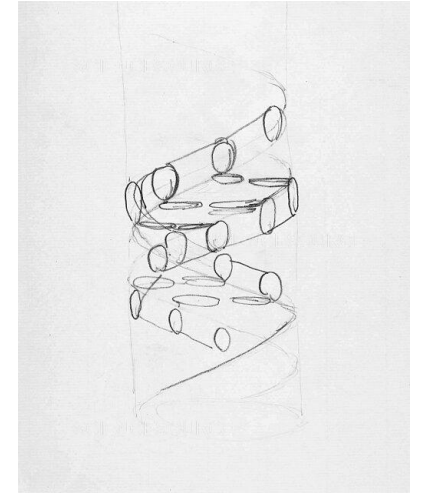
History of Nucleic Acids



<https://www.sciencesource.com/1694731-rosalind-franklin's-b-form-of-dna-1952.html>



<https://sciencehistory.org/education/scientific-biographies/james-watson-francis-crick-maurice-wilkins-and-rosalind-franklin/>



<https://www.sciencesource.com/2270297-crick's-original-dna-sketch.html>



FRIEDRICH
MIESCHER

Swiss Physician and
Biologist
First to isolate DNA in 1869
from the
Nuclei of white blood cells

Published: 25 April 1953

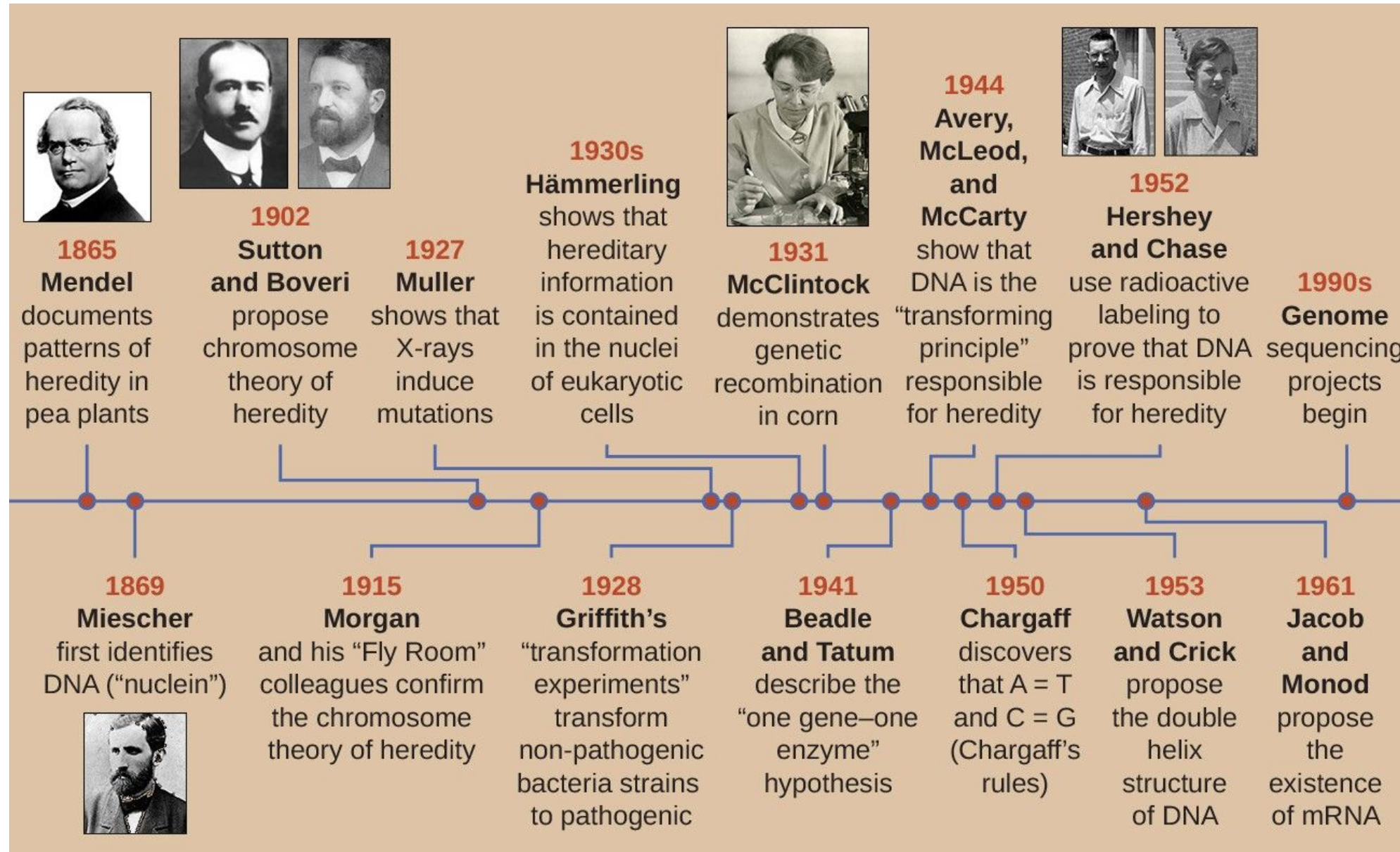
Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid

J. D. WATSON & F. H. C. CRICK

Nature **171**, 737–738 (1953) | Cite this article <https://doi.org/10.1038/171737a0>

An interesting story awaits-
https://www.pbs.org/wgbh/evolution/library/06/3/1_063_01.html

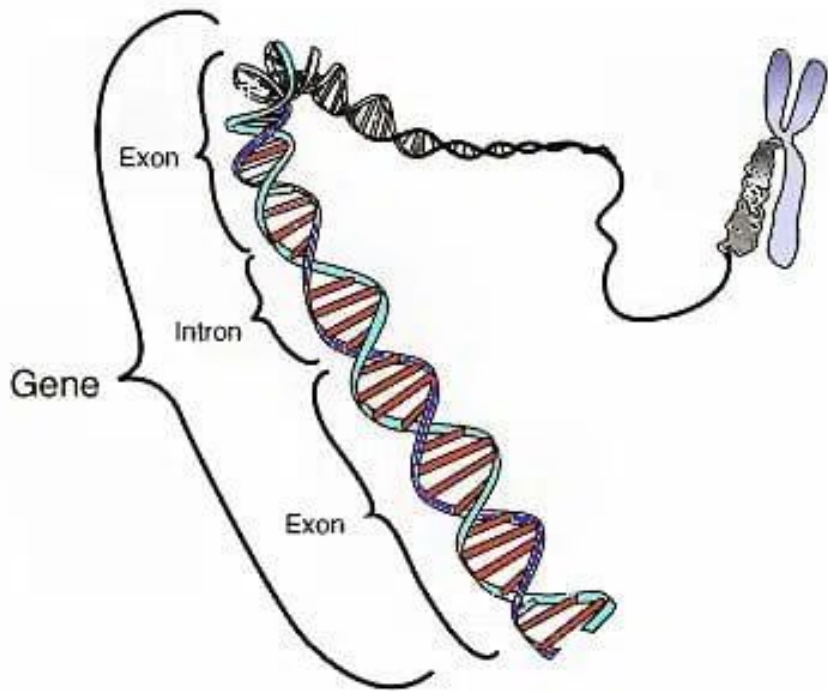
Major events



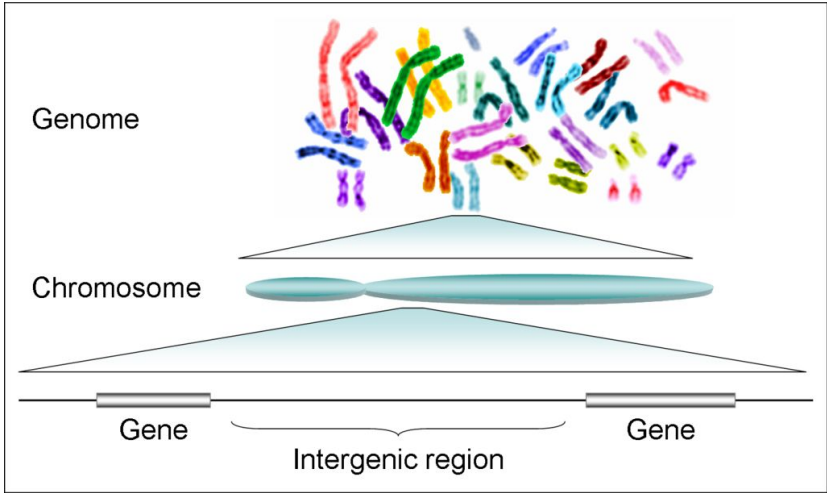
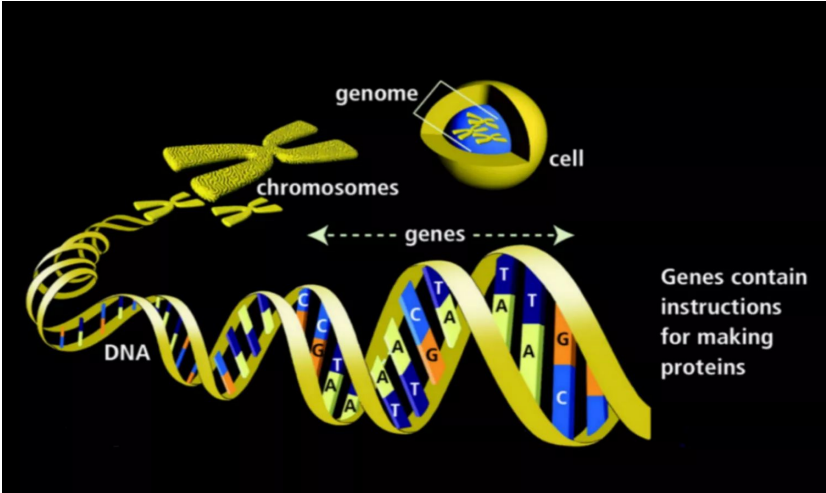
1965: For mRNA, Nobel Prize in Physiology and Medicine has been awarded

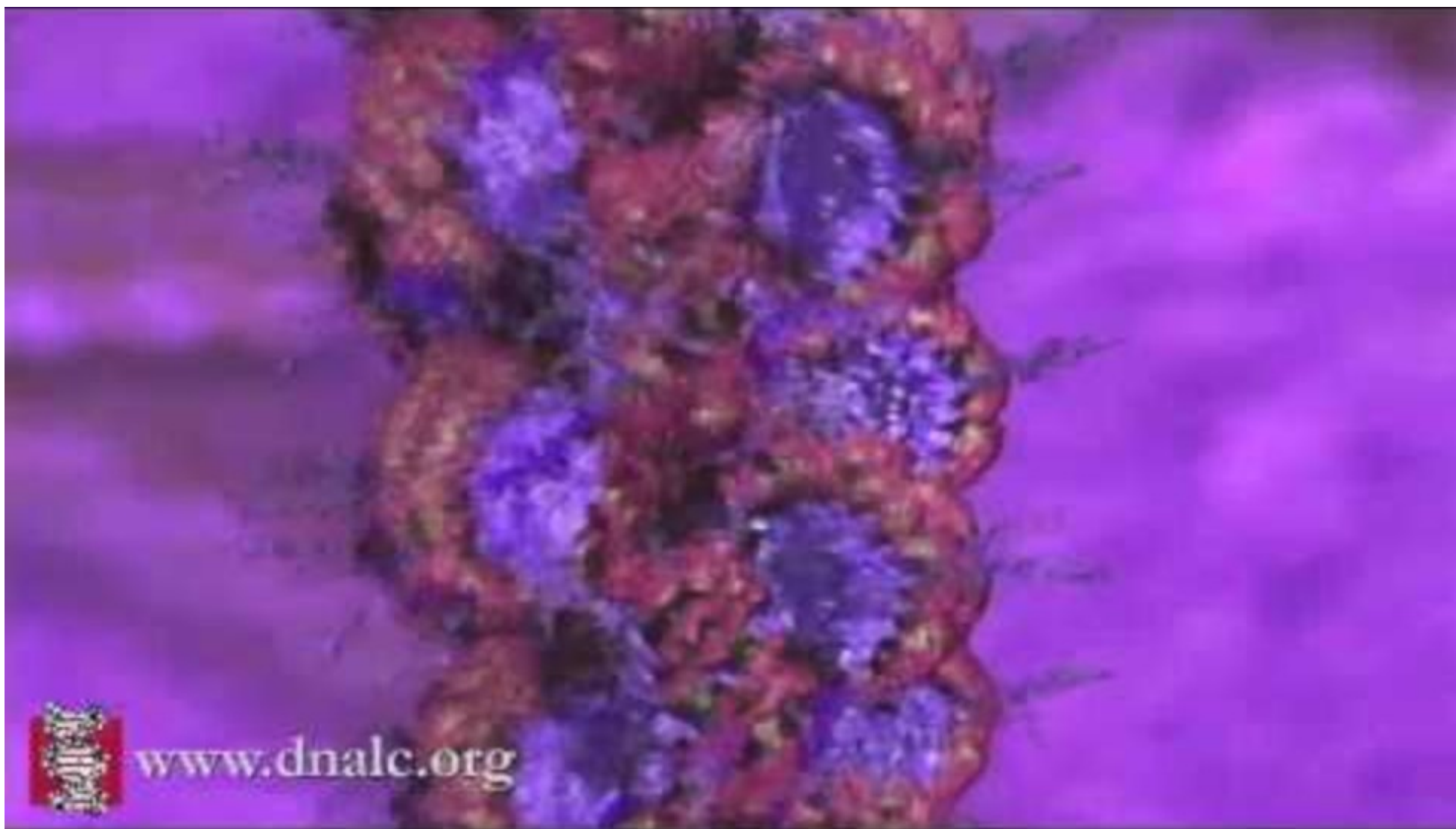
Genes and genomes

001



Organization of Eukaryotic Chromosomes	
DNA double helix	
DNA wrapped around histone	
Nucleosomes coiled into a chromatin fiber	
Further condensation of chromatin	
Duplicated chromosome	

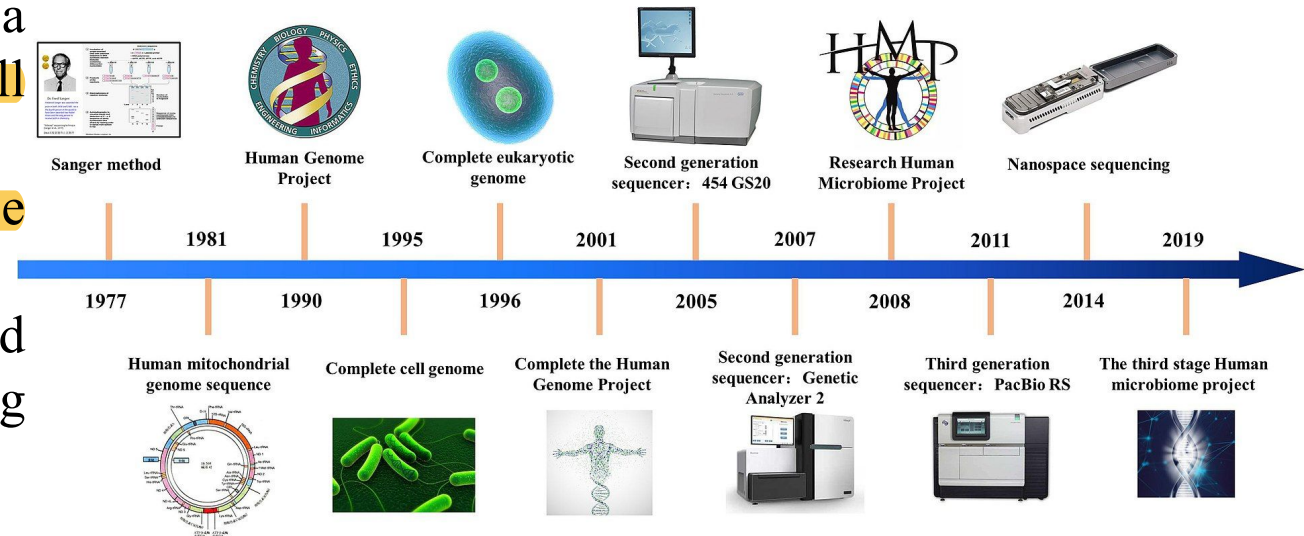
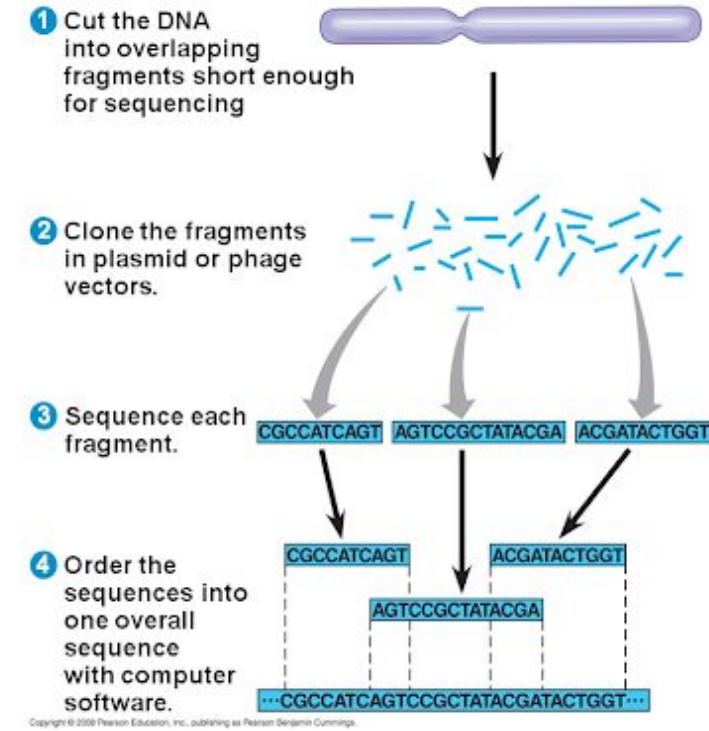


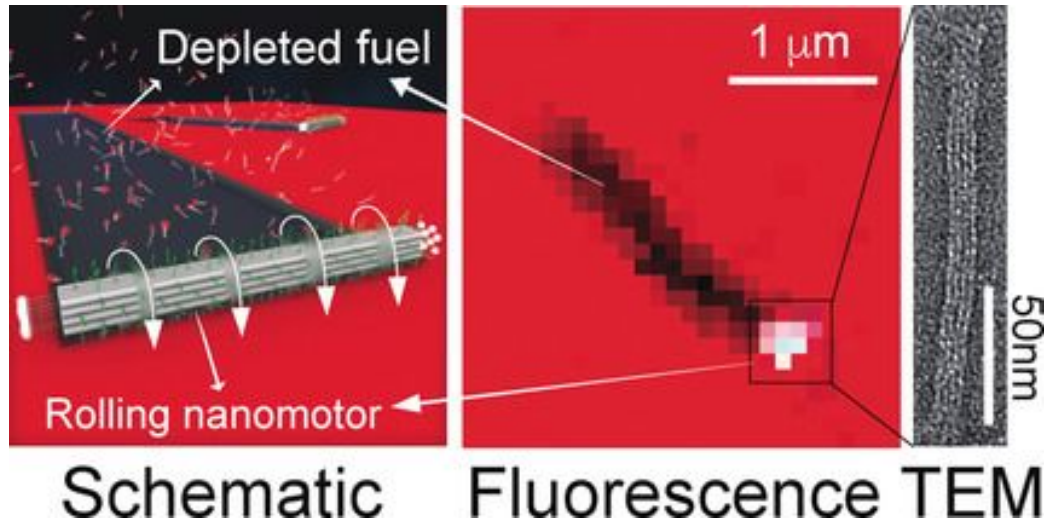


www.dnalc.org

Genome sequencing

- Process of identifying and determining the order of all the DNA base pairs (A, T, C, and G) in an organism's genome.
- It can be of different types-
 - **Whole Genome Sequencing (WGS):** WGS involves sequencing an organism's entire genome.
 - **Next-Generation Sequencing (NGS):** NGS technologies have greatly advanced genome sequencing by enabling high-throughput, cost-effective sequencing of DNA.
- After sequencing, the data must be assembled into a contiguous sequence, as it typically comes in **small fragments called "reads."**
- Then, the acquired data is matched with **Reference genomes.**
- Reference genomes are high-quality, well-annotated genome sequences used as a standard for comparing and aligning new sequencing data.

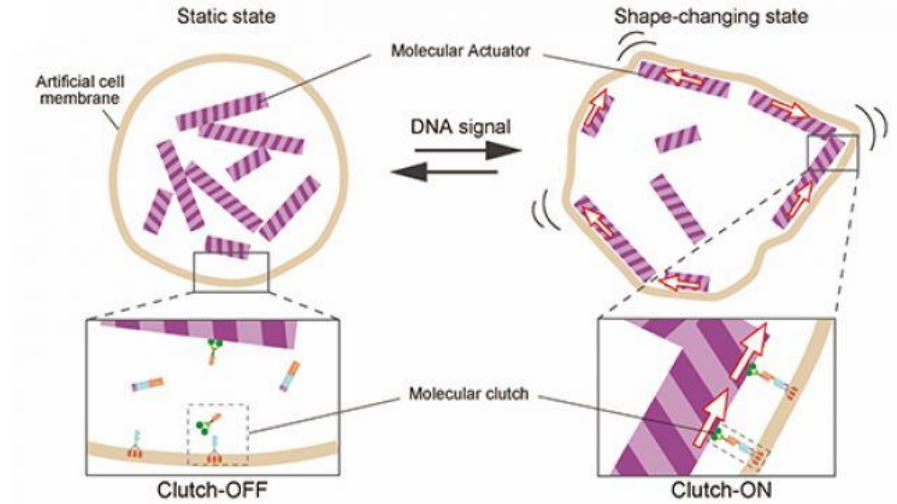




Tunable DNA Origami Motors Translocate Ballistically Over μm Distances at nm/s Speeds

Alisina Bazrafshan, Travis A. Meyer, Hanquan Su, Joshua M. Brockman, Aaron T. Blanchard, Selma Piranej, Yuxin Duan, Yonggang Ke✉, Khalid Salaita✉

First published: 03 February 2020 | <https://doi.org/10.1002/anie.201916281> | Citations: 36



Micrometer-sized molecular robot changes its shape in response to signal molecules

YUSUKE SATO , YUICHI HIRATSUKA , IBUKI KAWAMATA, SATOSHI MURATA, AND SHIN-ICHIRO M. NOMURA [Authors Info & Affiliations](#)

SCIENCE ROBOTICS • 1 Mar 2017 • Vol 2, Issue 4 • DOI: [10.1126/scirobotics.aal3735](https://doi.org/10.1126/scirobotics.aal3735)

More interesting stuff-

<https://doi.org/10.1021/acssynbio.0c00235>

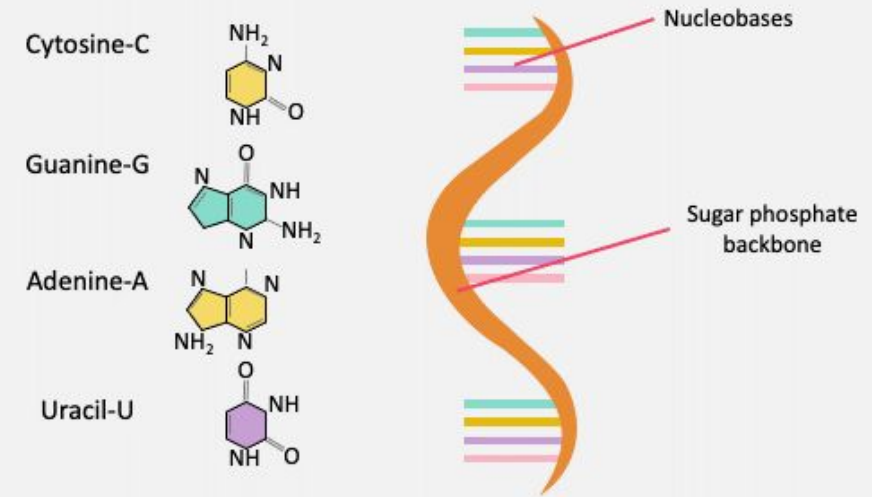
Ribonucleic Acid



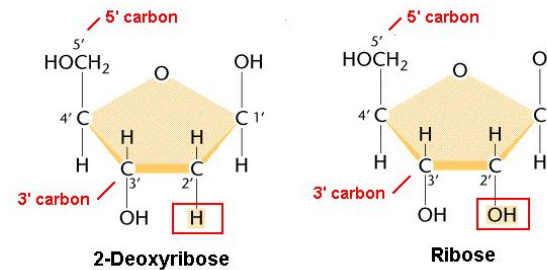
- Found in cells that plays a central role in various biological processes.
- Typically, **single-stranded**, but **can form secondary structures** which can be double stranded.
- RNA contains four nitrogenous bases: **adenine (A), uracil (U), guanine (G), and cytosine (C)**.
- RNA is involved in various processes related to genetic information transfer, including **transcription** (RNA synthesis from DNA), **reverse transcription** (RNA to DNA, as in retroviruses) and **translation** (from mRNA to protein.)
- Some RNAs are involved in **gene regulation** by block translation or lead to mRNA degradation.

RIBONUCLEIC ACID (RNA)

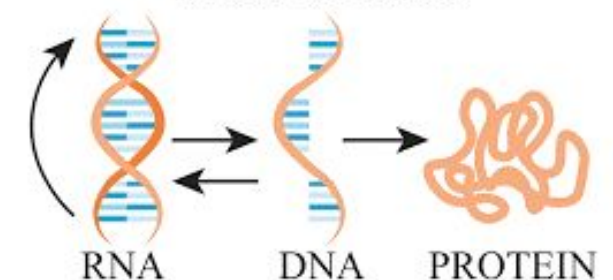
What is Ribonucleic Acid (RNA)



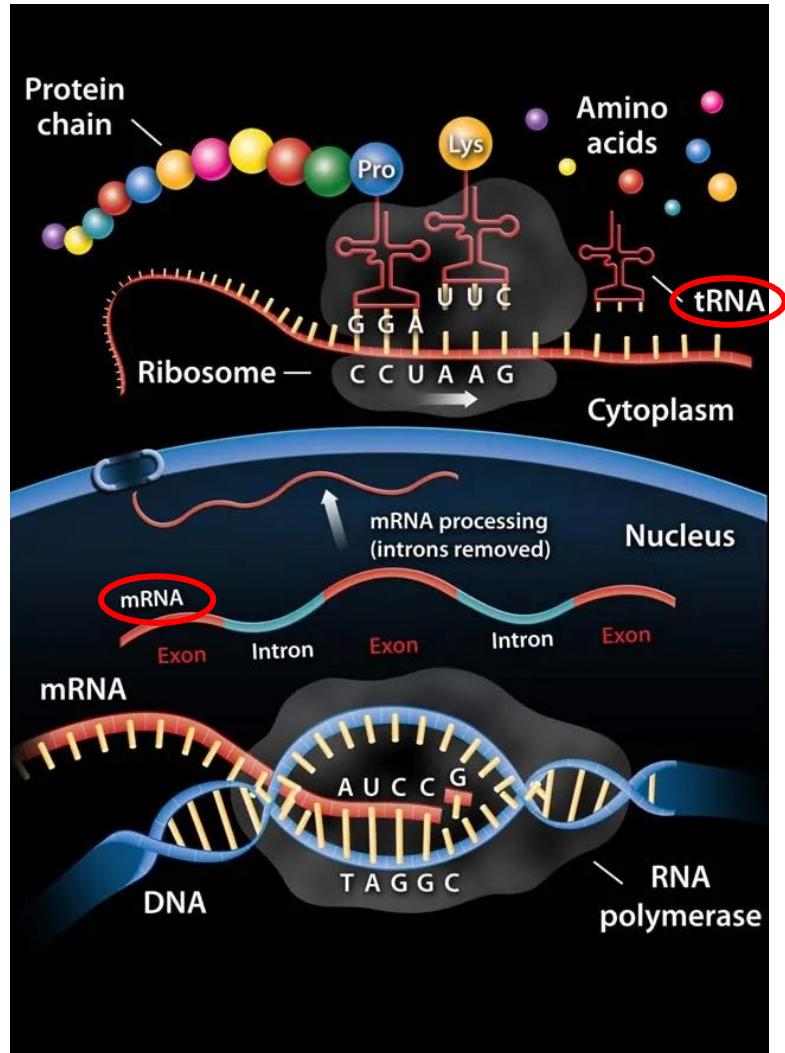
Ribonucleic acid (RNA) is a nucleic acid present in all living cells that has structural similarities to DNA. Unlike DNA, however, RNA is most often single-stranded. An RNA molecule has a backbone made of alternating phosphate groups and the sugar ribose, rather than the deoxyribose found in DNA



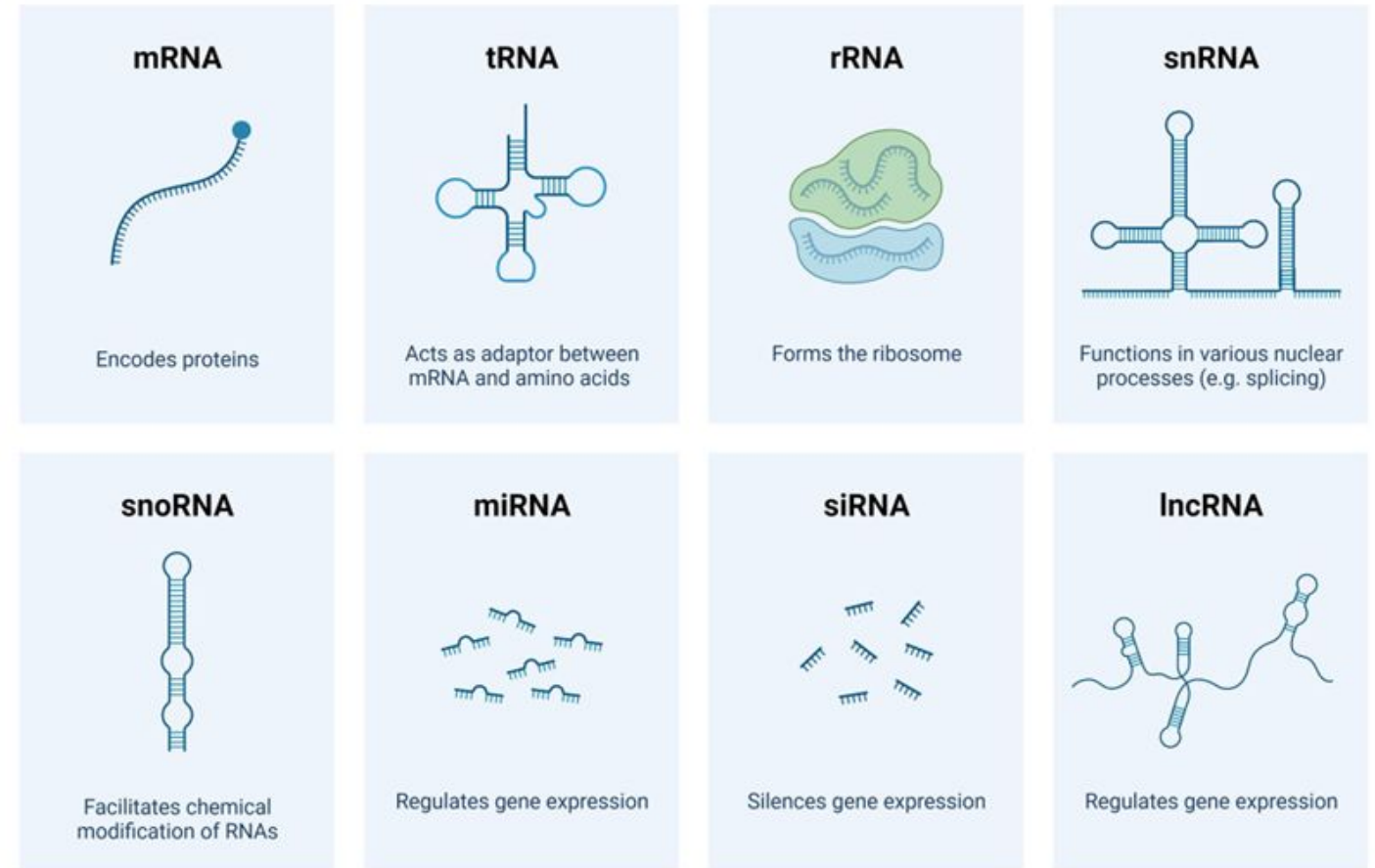
Role of RNA



Types of RNA



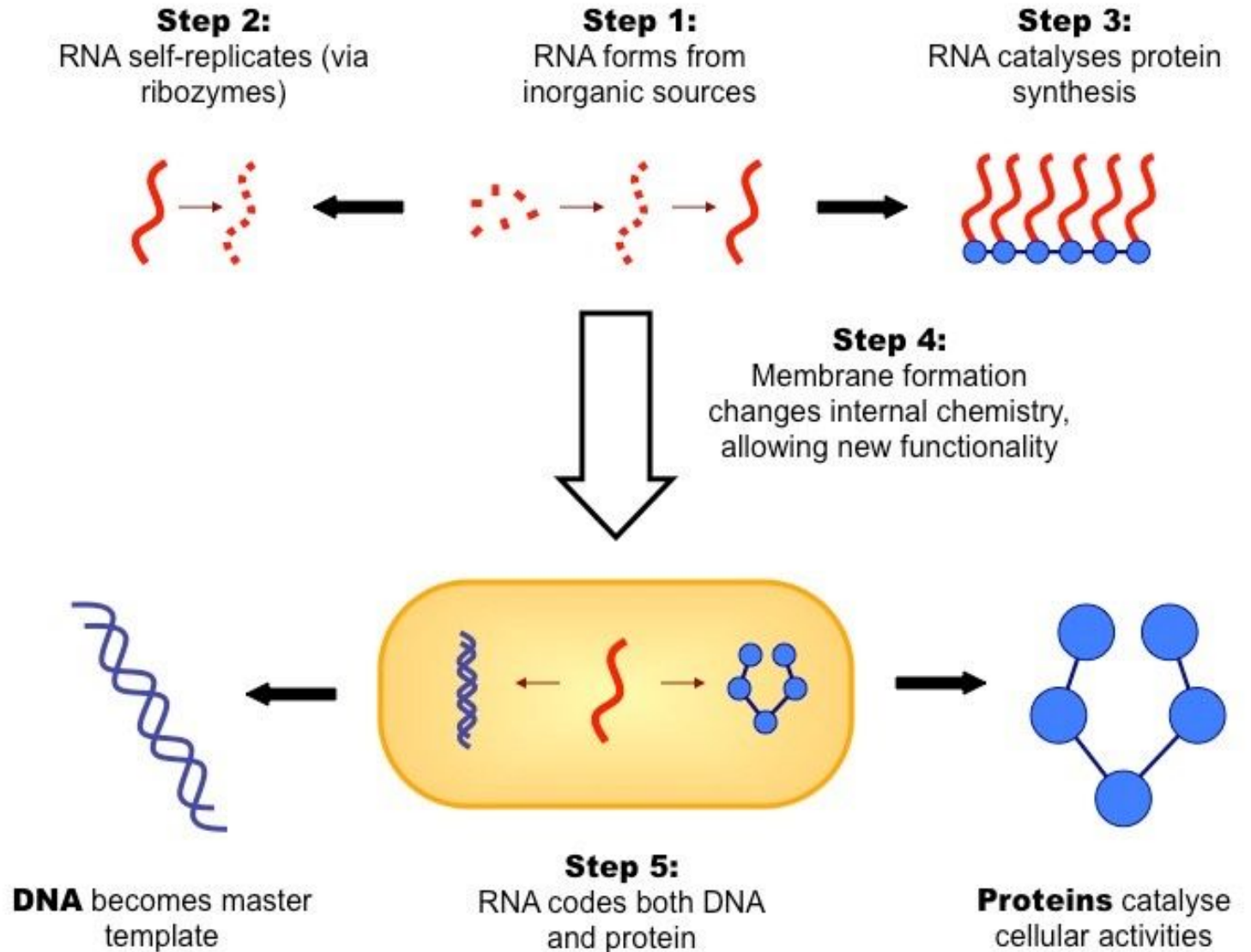
Types of RNA Produced in Cells



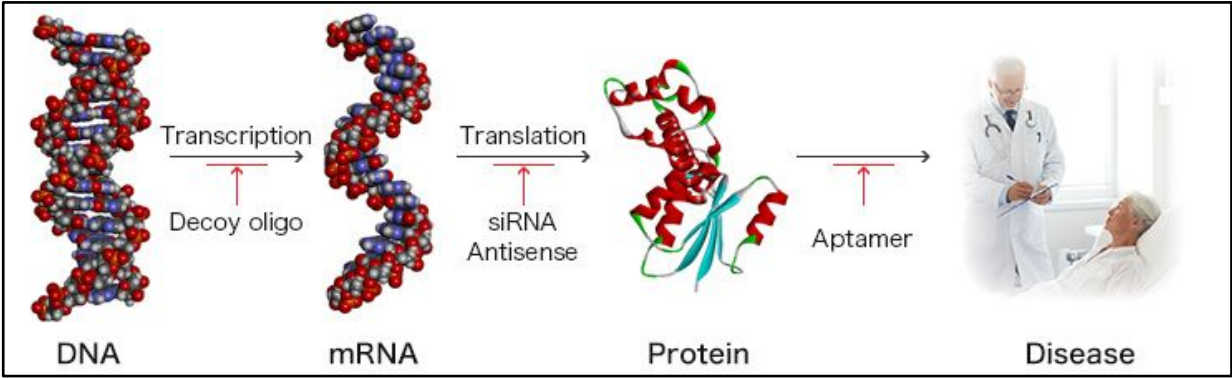
RNA world hypothesis

Before the emergence of DNA and proteins as the primary carriers of genetic information and catalysts for biological reactions, **RNA** (Ribonucleic Acid) played a **dual role** as both a **genetic molecule** and a **functional molecule**.

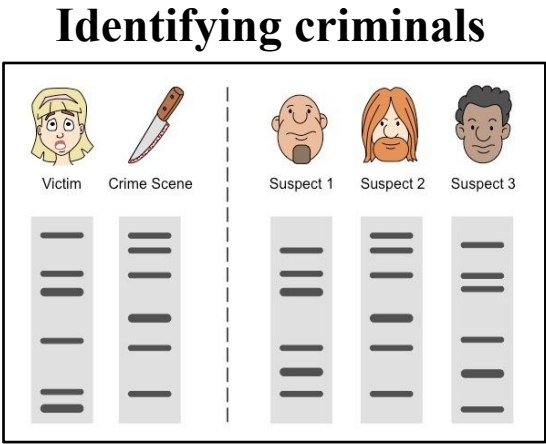
In the RNA world scenario, RNA molecules were **capable of storing genetic information** in their sequences, just like DNA does today. Simultaneously, RNA **exhibited catalytic properties**, acting as **ribozymes**, which are RNA molecules that can catalyze chemical reactions.



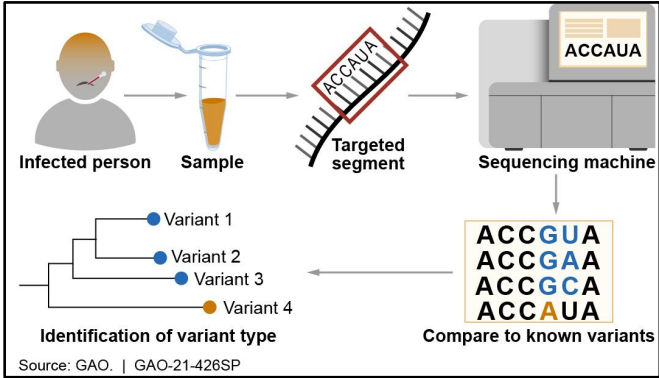
Applications



Gene knockdown therapy

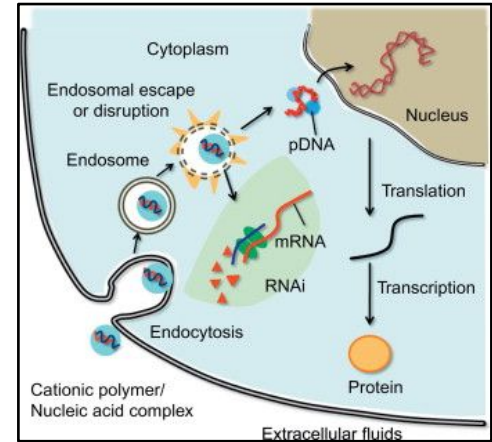


Identifying criminals

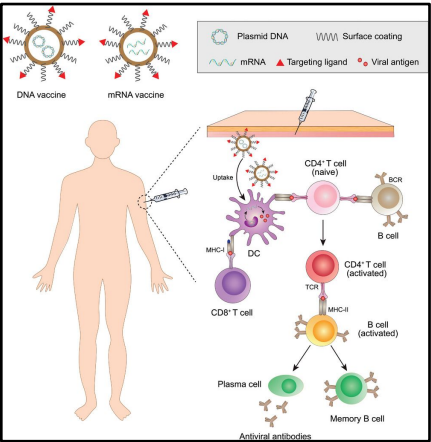


Diagnosis

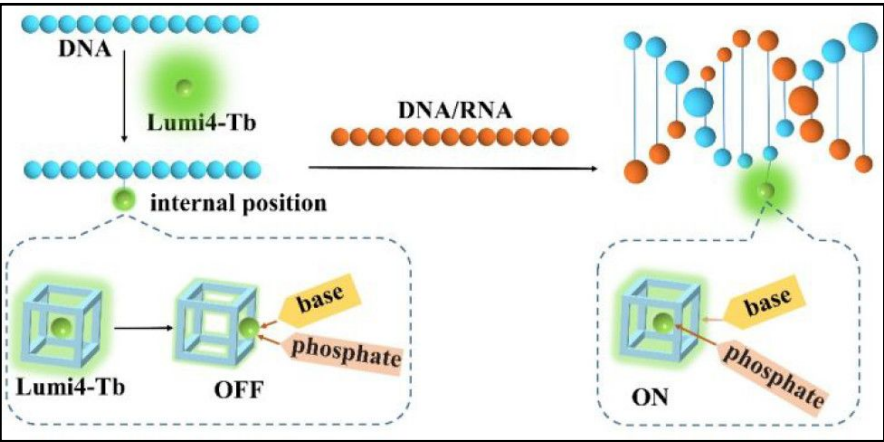
Nucleic acid applications



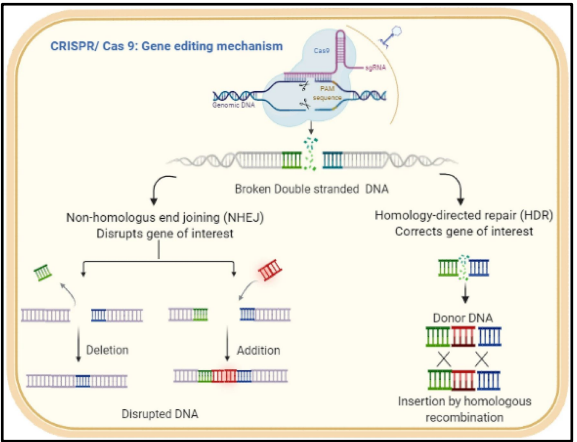
Drug delivery



DNA vaccine



Sensing



Gene Editing